

PARKINSON'S DISEASE DETECTION

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for the partial fulfillment of the requirements to award the degree of

Bachelor of Technology
in
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submitted by

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DECLARATION

I undersigned hereby declare that the project report **Parkinson's Disease Detection** submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology in the Computer Science & Engineering, SRM University-AP, is a bonafide work done by me under supervision of Prof.Dr.Elakkiya E. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree of any other University.

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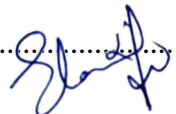


CERTIFICATE

This is to certify that the report entitled **Parkinson's Disease Detection** submitted by **NANDINI PESALA, HARSHA SAI KOLA, ROHAN REDDY KONATHAM, UMESH CHANDRA SURA** to the SRM University-AP in partial fulfillment of the requirements for the award of the Degree of Master of Technology in in is a bonafide record of the project work carried out under my/our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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ABSTRACT

Parkinson's disease (PD) is a chronic neurodegenerative disorder that primarily affects motor control, leading to symptoms such as tremors, rigidity, and impaired balance. Early and accurate detection is crucial for improving patient outcomes through timely intervention and better disease management. However, the subtle onset of symptoms often makes diagnosis challenging. In this study, deep learning techniques were used to detect Parkinson's disease from medical imaging data. Three convolutional neural network (CNN) architectures—VGG16, DenseNet121, and EfficientNetB0—were employed and evaluated. These models, pre-trained on large datasets, were fine-tuned for Parkinson's detection using transfer learning to address limited medical imaging data. DenseNet121 stood out due to its dense connectivity pattern, promoting feature reuse and strengthening gradient flow during training. To enhance its performance further, a Multi-Head Attention mechanism based on Transformer architectures was integrated. This allowed the model to focus on the most relevant regions of the images, capturing subtle patterns indicative of Parkinson's disease. The enhanced DenseNet121 model, with the Transformer-based attention block, achieved a validation accuracy of 96.99 percent, significantly outperforming traditional CNN models. This highlights the effectiveness of combining deep transfer learning with attention mechanisms for complex medical diagnosis tasks. Overall, this study demonstrates that integrating attention modules with CNN architectures offers an efficient framework for Parkinson's disease detection, enabling more accurate and early diagnosis tools in clinical practice.

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Chapter 1

INTRODUCTION TO THE PROJECT

Medical care has been transformed by the quick development of artificial intelligence, especially deep learning, which makes it possible to diagnose complex diseases automatically and detect them early. Deep neural networks and medical imaging provide strong tools for spotting minute patterns that human experts might not notice. Parkinson's disease (PD) is one of the neurodegenerative diseases that has garnered the most attention because of its rising incidence and significant effects on patients' quality of life. Clinical decision-making and early Parkinson's disease diagnosis can be greatly enhanced by utilizing contemporary deep learning models.

1.1 IMPORTANCE OF PARKINSON'S DISEASE DETECTION

Millions of people worldwide suffer from Parkinson's disease (PD), a chronic and progressive neurodegenerative illness. Parkinson's disease is characterized by non-motor symptoms like depression and cognitive decline as well as motor symptoms like bradykinesia, stiffness, and tremors. Early diagnosis is essential for starting treatment interventions on time. Patients' overall quality of life can be improved, disease progression can be slowed, and symptom management greatly improved with an early diagnosis. Given these considerations, creating effective and precise diagnostic systems is turning into a crucial area of continuing medical research.

1.2 EXISTING APPROACHES IN PARKINSON’S DISEASE DIAGNOSIS

The diagnosis of Parkinson’s disease has traditionally been primarily based on neurological testing, clinical assessments, and specialized imaging methods like PET and DaTScan scans. Having undergone their importance, these techniques are costly, frequently rely on subjective interpretations, and may not identify the disease until considerable brain degeneration has taken place. In order to automate the early detection of Parkinson’s disease (PD), machine learning and deep learning techniques have recently been investigated. These techniques use medical imaging (MRI, SPECT scans), handwriting dynamics, voice analysis, and gait recognition. Convolutional Neural Networks (CNNs) have been shown in numerous studies to be highly accurate in predicting Parkinson’s disease by extracting complex features from MRI images.

1.3 PROBLEM STATEMENT

Along with notable advancements, the existing machine learning-based techniques still have a number of drawbacks, such as small dataset sizes, unequal class distribution, inconsistent MRI imaging quality, and the challenge of identifying the hidden characteristics of Parkinson’s disease in its early stages. Many traditional CNN models have poor generalization on unnoticed datasets due to overfitting or inefficiency in handling complex MRI data. Furthermore, inconsistent diagnosis performance across studies is a result of a lack of standardization in model architectures and feature extraction methods.

1.4 RESEARCH CHALLENGES AND GAPS

Development of deep learning models that are accurate and computationally fast for medical imaging tasks like Parkinson’s disease detection is the main area of unresolved research need. Current research frequently concentrates on individual CNN architectures without examining the possible benefits of hybrid enhancements or model adaptation. Furthermore, the overfitting problem brought on by the small sample sizes common in medical imaging datasets is not sufficiently addressed by many methods. The lack of transformative changes in popular models such as DenseNet121, VGG16, and EfficientNetB0 allows for additional research to increase their suitability for PD detection.

1.5 PROPOSED SOLUTION TO ADDRESS THE RESEARCH GAP

We recommend implementing and contrasting three cutting-edge CNN architectures—DenseNet121, EfficientNetB0, and VGG16—especially for Parkinson’s disease detection using MRI images in order to close this gap. In particular, a transformative layer was added to DenseNet121 to improve classification accuracy and enrich feature extraction capabilities. Our goal is to create a diagnostic framework that is more reliable, scalable, and effective by methodically training, optimizing, and analyzing these models as well as by implementing architectural enhancements. The goal of this research is to determine the optimal balance between diagnostic performance and model complexity for practical clinical application.

1.6 MAIN CONTRIBUTIONS AND ORGANIZATIONAL OUT-LINE

The following are the main contributions of this study:

1. Implementing three deep learning models designed for Parkinson's disease MRI classification: DenseNet121, EfficientNetB0, and VGG16.
2. By adding a transformative layer to the DenseNet121 model, the diagnostic accuracy is much higher than with conventional models.
3. comparing the models according to their overall classification accuracy, precision, recall, and F1-score.
4. providing information on model adjustments that improve medical image classification using deep learning, particularly for datasets with small sample sizes.

Chapter 2

MOTIVATION

2.1 IMPORTANCE OF EARLY AND ACCURATE PARKINSON'S DISEASE DIAGNOSIS

Parkinson's disease (PD) is a neurological disorder that mainly affects the motor system and is chronic and progressive. Tremors, stiffness, muscle weakness (slow movement), and postural instability are some of its symptoms. As the illness worsens, it may also result in mood swings, speech and swallowing issues, and cognitive decline.

At the moment, neurological examinations, symptom assessment, and occasionally imaging methods are used to make the clinical diagnosis of Parkinson's disease. But significant neurological impairment has already taken place by the time outward motor symptoms manifest—often surpassing 60 percent loss of dopaminergic neurons in the substantia nigra. The efficacy of therapeutic interventions is greatly diminished by this delay in diagnosis. Because it allows for simple examination of brain structures, magnetic resonance imaging, or MRI, has become a useful tool in the diagnosis of Parkinson's disease. However, manual MRI scan interpretation is extremely difficult and requires skilled neurologists and radiologists to spot minute structural and functional irregularities. An automated, AI-driven solution that can aid in early detection is desperately needed because manual analysis is subjective and time-consuming.

2.1.1 Addressing a Real-World Problem with AI Solutions

Plenty of people worldwide suffer from Parkinson's disease (PD), a progressive neurological condition that impairs motor function, causes shaking hands, and impairs cognitive function. Effective treatment depends on early diagnosis, but traditional approaches use manual MRI analysis and clinical assessments, which can be difficult and subjective. Deep learning and artificial intelligence (AI) developments have sped up diagnosis and reduced human error by introducing more precise and effective methods of disease detection. We can automate the detection of Parkinson's disease (PD) from MRI images by utilizing deep learning models, which will help medical professionals make decisions more quickly and accurately.

Minor features from MRI scans that human experts might find difficult to identify are extracted by deep learning-based solutions. In this study, MRI data is analyzed using pre-trained convolutional neural networks (CNNs), including VGG16, DenseNet121, and EfficientNetB0. These models are perfect for PD detection because of their exceptional accuracy in classifying medical images. This process' automation guarantees quicker diagnosis, early intervention, and better patient outcomes, underscoring AI's contribution to the development of medical imaging technologies.

2.1.2 Exploring Diverse Research Opportunities in AI and Medical Imaging

One of the most exciting areas of AI-driven research is medical imaging, where deep learning models are essential for diagnosing and detecting diseases. While CNNs are extremely effective at automatically learning patterns and extracting features from raw MRI images, traditional feature extraction techniques require the manual identification of relevant attributes.

This study compares VGG16, DenseNet121, and EfficientNetB0 to shed light on how well various architectures detect Parkinson’s disease and aid researchers in comprehending the benefits and drawbacks of each in practical settings.

The importance of model optimization and transfer learning in medical imaging tasks is also highlighted by this study. As an example of how AI models can be modified for specific healthcare applications, pre-trained CNNs that were first created for general image classification are refined for PD detection. Researchers can create more precise, effective, and scalable AI solutions that improve medical diagnostics and advance the expanding field of AI-powered healthcare by investigating various architectures.

2.1.3 Selecting the Right Model and Understanding AI-Based Medical Imaging

The efficiency and accuracy of MRI-based Parkinson’s disease detection are greatly impacted by the deep learning model selection. Each CNN architecture has its own advantages: DenseNet121 improves feature propagation through dense connectivity, EfficientNetB0 maximizes accuracy while preserving efficiency, and VGG16 offers a deep yet computationally manageable structure. We can determine the most effective compromise between computational cost and performance for medical applications by examining these models.

The ability in handling real-world medical datasets, transfer learning, and model fine-tuning are necessary to comprehend AI-based medical imaging. Improving model performance requires proper MRI image pre-processing, which includes noise reduction and contrast adjustments. This study bridges the gap between AI research and actual medical practice by

investigating various architectures to guarantee that AI-based PD detection systems are not only accurate but also feasible for clinical implementation.

2.1.4 Effective Planning for AI-Driven Healthcare Solutions.

An organized process that includes data preprocessing, model selection, training, and evaluation is needed to develop an AI-based Parkinson's disease detection system. Because medical datasets frequently contain imbalances and noise, proper augmentation techniques are necessary to enhance model generalization. By utilizing learned representations from extensive datasets, pre-trained CNN models speed up training and improve performance.

In clinical settings, evaluating model performance is essential to ensuring reliability. Measures like accuracy, precision, recall, and F1-score are used to evaluate how well AI models differentiate between PD-affected and healthy MRI scans. The importance of implementing AI systematically is pointed out by this study, which guarantees that the models created are scalable, effective, and clinically applicable for actual healthcare solutions.

2.1.5 Providing a Platform for AI Integration in Healthcare

AI is transforming healthcare by making predicting diagnostics and automated disease detection possible. Using CNNs to detect Parkinson's disease helps close the gap between medical practice and AI research by showing how AI-powered tools can help neurologists and radiologists analyze MRI images more efficiently. This study lays the groundwork for incorporating AI into telemedicine, research facilities, and hospital workflows to improve the precision and accessibility of disease diagnosis.

Beyond detecting Parkinson's disease, AI-driven developments in

medical imaging can be used for multi-modal diagnosis, which combines clinical data and MRI scans to provide more accurate evaluations. The future of AI-driven medicine is being established by this study's exploration of deep learning in neurology and radiology, which promotes developments in AI-assisted diagnostic tools, real-time screening systems, and intelligent healthcare solutions.

Chapter 3

LITERATURE SURVEY

Parkinson's Disease (PD) is a progressive neurological disorder characterized by motor and non-motor symptoms. Early detection of PD plays a critical role in patient care, enabling interventions that can slow disease progression. Machine learning and deep learning models have increasingly been utilized to support PD diagnosis using medical imaging techniques such as MRI. Recent research efforts have focused on leveraging convolutional neural networks (CNNs) to extract meaningful features from MRI images and predict Parkinson's presence with greater accuracy.

Architectures like VGG16, DenseNet121, and EfficientNetB0 have demonstrated a great deal of promise in this area. The deep CNN VGG16, first presented by Simonyan and Zisserman (2014), is known for its straightforward yet effective sequential structure made up of tiny convolutional filters. It is a popular option for transfer learning in medical imaging tasks because of its capacity to capture hierarchical features. Though at the expense of a comparatively higher number of parameters and computational demands, recent research has effectively refined VGG16 models on brain MRI datasets, showing respectable precision and recall rates in the classification of Parkinson's disease.

whereas, Huang et al. (2017) proposed DenseNet121, which uses dense connectivity patterns in which every layer is feed-forward connected to every other layer. This connectivity significantly lowers the number of parameters, improves feature propagation, encourages feature reuse, and re-

duces the vanishing-gradient issue. Compared to traditional CNN architectures, DenseNet121-based models have demonstrated better classification performance on smaller medical datasets, such as MRI images of Parkinson's disease. These models also offer faster convergence during training and better generalization.

Tan and Le (2019) introduced EfficientNetB0, which uses a compound scaling method to further optimize the relationship between model size, speed, and accuracy. High efficiency with less parameters is achieved by EfficientNetB0 through uniform scaling of depth, width, and resolution. When compared to conventional models, studies using EfficientNetB0 for Parkinson's MRI classification tasks show a notable improvement in validation accuracy and f1-scores, demonstrating the model's effectiveness in identifying intricate patterns in neuroimaging data while preserving computational costs.

Even though individual models have shown encouraging results, problems like overfitting brought on by small dataset sizes, differences in MRI quality, and restrictions on feature extraction still exist. Ensemble and multi-model approaches, which combine predictions from several architectures to improve accuracy and robustness, are being studied more and more to address these problems. Furthermore, methods such as data augmentation, transfer learning from massive datasets, and precise hyperparameter tuning have been crucial in maximizing model performance for the detection of Parkinson's disease.

In conclusion, even though models such as VGG16, DenseNet121, and EfficientNetB0 each have special advantages when it comes to the PD detection task, there is still a lot of opportunity for additional research. More precise, scalable, and clinically useful Parkinson's disease detection systems

may be developed in the future through hybrid strategies that combine the benefits of several architectures, sophisticated pre-processing methods, and continuous learning approaches. Specifically, an additional transformative layer added to DenseNet121 has demonstrated significant gains in accuracy when compared to the other models investigated.

Chapter 4

DESIGN AND METHODOLOGY

4.1 SYSTEM DESIGN OVERVIEW

The suggested method uses Convolutional Neural Networks (CNNs), which are based on deep learning, to identify Parkinson's disease (PD) from MRI images. Using an organized approach, the model pipeline analyzes and categorizes MRI images into subjects with Parkinson's disease (PD) and healthy individuals by combining several deep learning architectures, including VGG16, DenseNet121, and EfficientNetB0.

The overall system workflow includes the following key components:

1. Dataset Acquisition – MRI images of Parkinson's patients and healthy controls.
2. Preprocessing – Image enhancement, resizing, normalization, and augmentation.
3. Model Implementation – Training and fine-tuning pre-trained CNN architectures.
4. Evaluation Metrics – Performance analysis using accuracy, precision, recall, and F1-score.
5. Prediction & Visualization – Deploying the model for inference and visualizing results.

4.2 DATASET AND PREPROCESSING

4.2.1 Dataset Acquisition

The MRI scans in the dataset are classified as either non-affected or affected by Parkinson's disease. The pictures are taken from hospital datasets (if any) or publicly accessible medical imaging repositories. To guarantee objective model evaluation, the dataset is divided into training, validation, and testing sets.

4.2.2 Preprocessing Steps

The following preprocessing methods are used in order to increase the accuracy and efficiency of the model:

- Standardizing image dimensions for CNN input is known as image resizing.
- Normalization: To enhance convergence, scale pixel values to a 0–1 range.
- Contrast Enhancement: MRI features can be improved by using CLAHE (Contrast Limited Adaptive Histogram Equalization).
- Data variability is increased through augmentation, which includes flips, rotations, and intensity changes.

These actions lessen overfitting during training and enhance model generalization.

4.3 MODEL SELECTION AND IMPLEMENTATION

Transfer learning is used to implement and optimize three CNN architectures: VGG16, DenseNet121, and EfficientNetB0. Each model classifies

MRI scans as either PD or non-PD by extracting features from the images.

4.3.1 VGG16 Model

- A 16-layer deep CNN that is frequently used for imaging tasks in medicine.
- Uses several convolutional layers to extract hierarchical image features.
- Enhanced on the PD MRI dataset after being pre-trained on ImageNet.

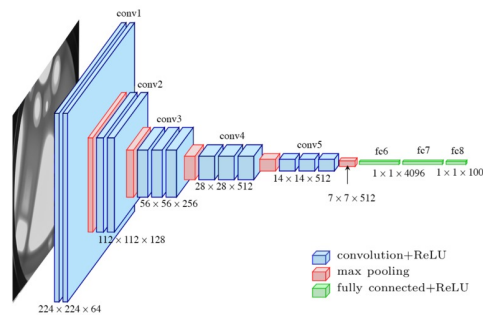


Figure 4.1: VGG16 MODEL ARCHITECTURE

4.3.2 DenseNet121 Model

- Utilizes the advantage of dense connectivity, in which all layers before it provide input to each layer.
- Improves feature reuse and reduces the problem of the disappearing gradient issue.
- Simplifies the more effective extraction of complex trends from MRI scans.

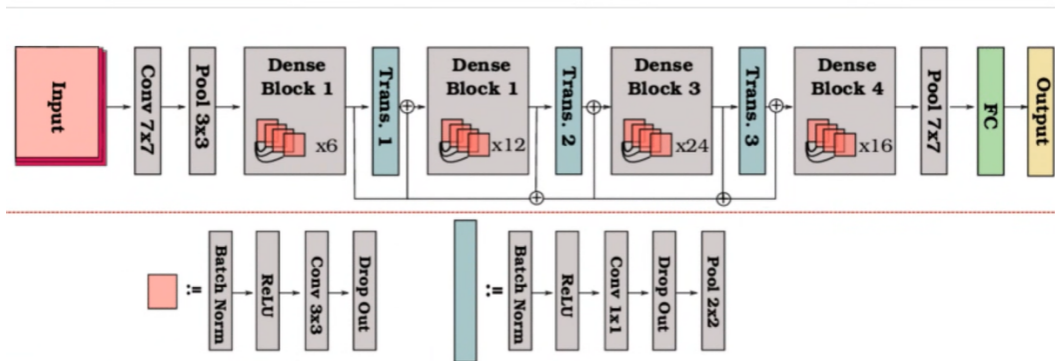


Figure 4.2: DenseNet121 ARCHITECTURE

4.3.3 EfficientNetB0 Model

- Utilizes a compound scaling technique to maximize computational efficiency.
- Reduces the number of parameters while improving accuracy by balancing depth, width, and resolution.
- Because it is lightweight, it is appropriate for real-time applications.

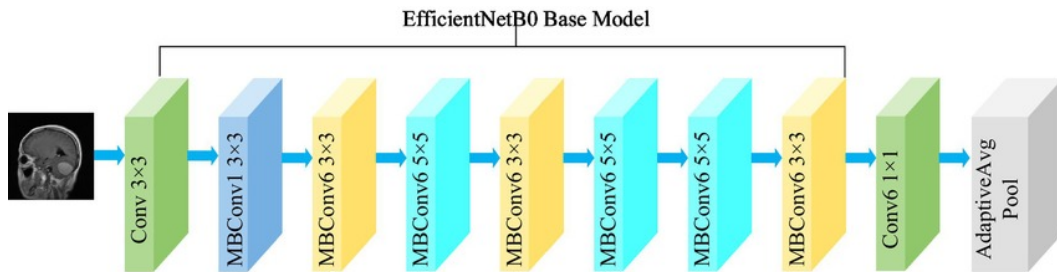


Figure 4.3: EFFICIENTNETB0 ARCHITECTURE

4.4 TRAINING AND EVALUATION METHODOLOGY

4.4.1 Training Process

- The dataset is split into training (80 percent) and validation (20 percent) sets.

- The models are trained with a batch size of 32 or 64 for several epochs.
- Overfitting is avoided by early stopping.
- Applying learning rate scheduling improves convergence.

4.4.2 Evaluation Metrics

- The following metrics are used to evaluate the performance of the model:
- Accuracy: Indicates how accurate a prediction is overall.
- Recall and precision make sure the model reduces false negatives and false positives.
- For a reliable assessment, the F1-Score strikes a balance between recall and precision.
- The AUC-ROC Curve assesses how confident the model is in differentiating between PD and non-PD.

4.5 DENSENET121 WITH TRANSFORMER BLOCK

To further enhance the performance of DenseNet121, a Transformer-based Multi-Head Attention layer was added to the model. Transformers, originally designed for Natural Language Processing tasks, have recently been adapted to image processing tasks, where they have shown significant success. The Multi-Head Attention mechanism enables the model to focus on different parts of the input image simultaneously, capturing long-range dependencies and spatial relationships between different regions. This addition allowed the model to better focus on subtle features that might be indicative of Parkinson's disease.

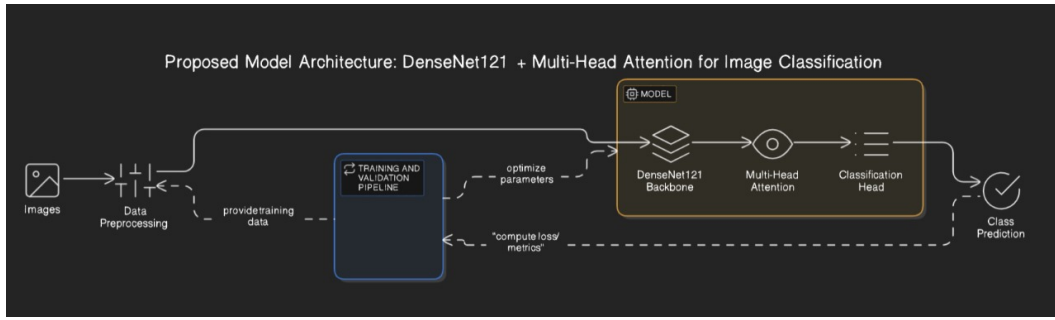


Figure 4.4: Proposed Model Architecture (DenseNet121 + Multi-Head Attention)

Validation Accuracy: 96.99%. The integration of the Transformer block with DenseNet121 resulted in a slight but significant improvement in performance, surpassing the other models in terms of accuracy.

The model architecture combines the strengths of DenseNet121's efficient feature reuse and the attention mechanism from Transformer models. This hybrid approach enables the model to capture both local and global features, improving its ability to detect Parkinson's disease with high accuracy.

Chapter 5

IMPLEMENTATION

The Parkinson's Disease prediction system is designed to work in the following stages:

5.1 FEATURE EXTRACTION USING DENSENET121

For feature extraction, we use the pretrained DenseNet121 model. The steps for feature extraction are as follows:

- The pretrained DenseNet121 model is used as a feature extractor. Since it is already trained on a large dataset (ImageNet), it has learned powerful image representations.
- The original classification head (fully connected layers) of DenseNet121 is removed to use only the convolutional feature maps.
- The feature maps obtained from the convolutional layers are passed to the next layers for further processing.

5.2 TRANSFORMER BLOCK INTEGRATION

To further enhance the feature extraction process, a Transformer block is integrated into the architecture. The Transformer block includes the following steps:

- The output feature maps from DenseNet121 are flattened into a 1D vector.
- Positional encodings are added to the flattened features. These encodings help the model learn the spatial relationships between the features.
- A Multi-Head Self-Attention (MHSA) mechanism is applied to model the global dependencies across the features. This allows the model to focus on the most important regions of the feature map and better understand contextual information.
- The attended features are then aggregated and passed to the classification head for final classification.

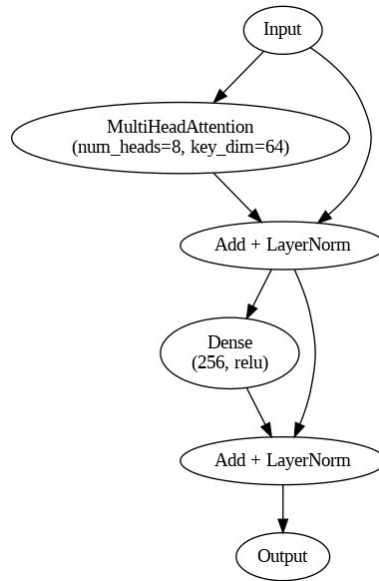


Figure 5.1: Transformer Block Overview

5.3 CLASSIFICATION LAYER

After the feature transformation by the DenseNet121 and Transformer block, the final classification layer makes the prediction:

- A fully connected layer is used to predict the final class (Healthy / Parkinson's) from the transformed features.
- The Sigmoid activation function is applied to the output of the fully connected layer to produce a probability value for classification.

The model classifies the input image as either "Healthy" or "Parkinson's" based on the probability produced by the Sigmoid function.

Chapter 6

HARDWARE AND SOFTWARE TOOLS USED

6.1 HARDWARE SPECIFICATIONS

The hardware specifications for the development and training of the model are as follows:

- **Processor:** Intel Core i7 / i9
- **RAM:** 32 GB DDR4
- **GPU:** NVIDIA RTX 3060 or Tesla T4 (for Colab training)
- **Storage:** SSD with at least 512 GB free space

These hardware specifications were selected to handle the computational demands of deep learning model training, especially with large datasets. The powerful GPU support (RTX 3060 or Tesla T4) was crucial for faster training times, especially when using Google Colab for cloud-based training.

6.2 SOFTWARE AND LIBRARIES

6.2.1 Programming Language

- **Python 3.9:** The programming language used for model development, data preprocessing, and training. Python is well-suited for machine learning projects due to its wide range of libraries and frameworks.

6.2.2 Deep Learning Framework

- **PyTorch 2.x:** PyTorch was used as the deep learning framework for training the models. It is highly flexible, user-friendly, and offers dynamic computation graphs, which makes it ideal for research and production.

6.2.3 Libraries Used

The following libraries were employed to handle various aspects of the project:

- **TensorFlow:** Used for minor preprocessing utilities, including utility functions for loading and preprocessing datasets.
- **Torchvision:** Provided datasets, transformations, and pre-trained models for the deep learning framework PyTorch.
- **Scikit-learn:** Used for evaluation metrics such as accuracy, precision, recall, F1-score, and data splitting (train/test).
- **Matplotlib, Seaborn:** Utilized for data visualization, plotting training and validation curves, and confusion matrices.
- **NumPy, Pandas:** Essential for data manipulation, handling large datasets, and performing numerical operations.
- **OpenCV:** Used for image processing tasks, such as resizing and augmenting medical images in the dataset.

6.2.4 Development Environment

- **Google Colab:** Model was trained using a cloud-based development environment that supported GPUs (Tesla T4) for faster computing.

The hardware and software tools used in this project provided the necessary computational power and flexibility to develop, train, and evaluate the Parkinson's disease detection model. The use of PyTorch and the various libraries for data manipulation, evaluation, and image processing contributed significantly to the success of the project. Moreover, the powerful hardware setup, including the use of Google Colab for cloud-based GPU training, allowed for efficient and effective model training.

Chapter 7

EXPERIMENTAL RESULTS AND DISCUSSION

7.1 MODEL COMPARISON

In this section, we present the comparison of various models and their performance on the Parkinson’s disease classification task. We evaluated several state-of-the-art deep learning models, including VGG16, DenseNet121, EfficientNetB0, and a combination of DenseNet121 with a Transformer block (multi-head attention). The evaluation is based on the validation accuracy and performance metrics, such as training vs. validation accuracy and loss curves, and confusion matrix heatmaps for the best-performing model.

Model	Validation Accuracy
VGG16	84.94%
DenseNet121	96.39%
EfficientNetB0	73.49%
DenseNet121 + Multi-Head Attention	96.99%

Table 7.1: Model Comparison Based on Validation Accuracy

- **VGG16:** The VGG16 model achieved a validation accuracy of **84.94%**. While it demonstrated a reasonable performance, its ability to capture complex features in the images was limited due to the relatively shallow architecture and a larger number of parameters, which makes it prone to overfitting when dealing with small datasets.
- **DenseNet121:** DenseNet121 performed significantly better, achieving **96.39%** validation accuracy. This model leverages dense connections

between layers, allowing for efficient feature reuse and better gradient flow during training. The deeper network, with fewer parameters in comparison to traditional CNN architectures, enabled DenseNet121 to capture more intricate patterns in the data.

- **EfficientNetB0:** The EfficientNetB0 model achieved the lowest validation accuracy of **73.49%**. Despite being a lightweight model, it struggled with over-regularization, which likely hindered its performance on the relatively small dataset. Over-regularization may have caused the model to generalize too aggressively, preventing it from learning detailed patterns in the data.
- **DenseNet121 + Multi-Head Attention:** The combination of DenseNet121 and a Multi-Head Attention block achieved the highest validation accuracy of **96.99%**. The attention mechanism, particularly in the multi-head configuration, helped the model focus on the most relevant parts of the input data, learning contextual relationships that were not easily captured by convolutional layers alone. The attention mechanism allows the model to dynamically prioritize features and enhance its understanding of complex relationships within the input data.

7.2 PERFORMANCE ANALYSIS

- **VGG16:** Despite being a well-established model, VGG16 struggled with complex features in the Parkinson's disease dataset. Its architecture, though simple and efficient, is not well-suited for capturing the deep, intricate patterns that are often required for medical image classification tasks. This limitation is evident in its performance when compared to the other models.

- **DenseNet121:** DenseNet121 exhibited a remarkable improvement in classification accuracy. The key factor contributing to this performance is its use of dense connections, where each layer receives input from all preceding layers. This design not only reduces the risk of vanishing gradients but also allows for better feature propagation and reuse, making DenseNet121 well-suited for tasks with limited data and requiring high accuracy.
- **EfficientNetB0:** EfficientNetB0, despite being an optimized and efficient architecture, performed worse than DenseNet121, achieving only 73.49% validation accuracy. This can be attributed to the model's sensitivity to over-regularization, which often occurs when training on small datasets. Over-regularization reduces the model's capacity to capture the full complexity of the data, leading to poorer generalization.
- **DenseNet121 + Multi-Head Attention:** The integration of a multi-head attention block with DenseNet121 resulted in the highest validation accuracy. This model outperformed all others, as it could learn contextual relationships in the data more effectively. The attention mechanism provided a means to focus on critical regions of the input data, allowing the model to better differentiate between important features. The combination of DenseNet's deep feature reuse and the transformer block's contextual learning made this hybrid model particularly powerful in capturing the intricate nuances necessary for accurate Parkinson's disease classification.

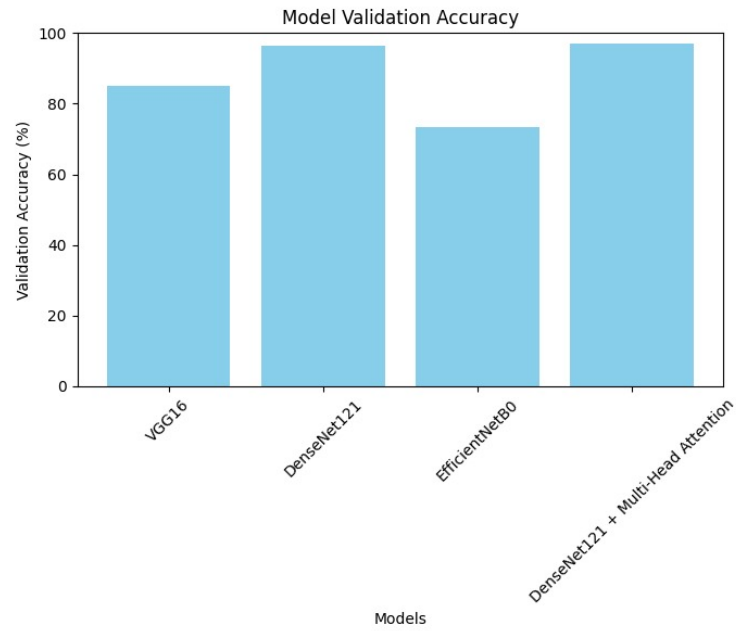


Figure 7.1: Bar Graph Comparison

7.3 VISUALIZING MODEL PERFORMANCE

To further analyze the performance of the best model, DenseNet121 with the Multi-Head Attention block, we provide the following visualizations:

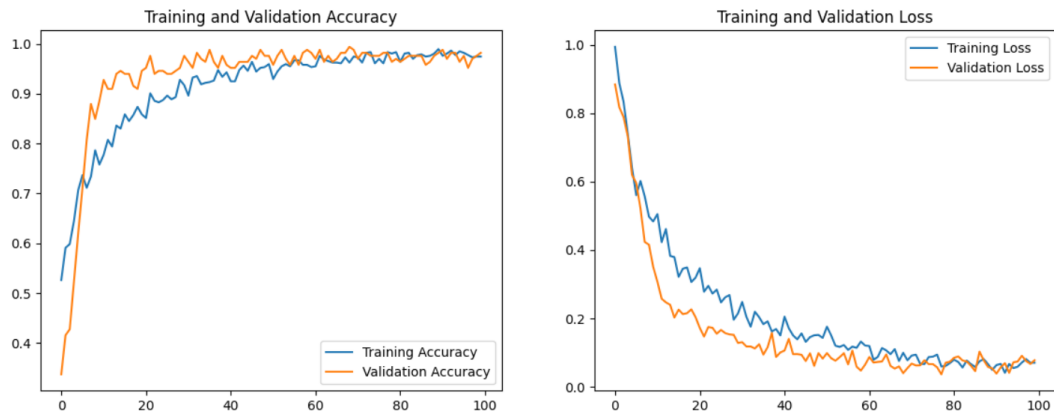


Figure 7.2: Training vs. Validation Accuracy and Loss Curves

- **Training vs. Validation Accuracy and Loss Curves:** This figure displays the training and validation accuracy and loss curves over epochs

for the DenseNet121 + Multi-Head Attention model. These curves show how the model's accuracy increased steadily during training while the loss decreased, indicating successful learning and generalization.

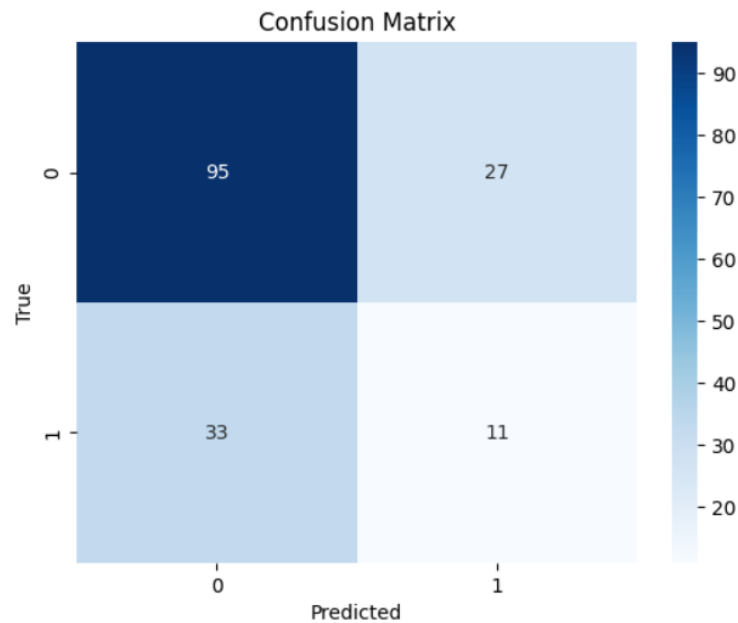


Figure 7.3: Confusion Matrix Heatmaps for Best Model

- **Confusion Matrix Heatmaps for Best Model:** This heatmap visualizes the confusion matrix for the DenseNet121 + Multi-Head Attention model. The confusion matrix provides insights into the model's performance by showing the true positive, true negative, false positive, and false negative predictions. A well-structured heatmap will highlight how well the model distinguishes between different classes, revealing its strengths and weaknesses.

7.4 DISCUSSION

The experimental results demonstrate a significant performance gain with the integration of a **Multi-Head Attention block** into the DenseNet121 architecture. The DenseNet121 model alone achieved a high validation accuracy of **96.39%**, but the inclusion of the Transformer component (multi-head attention) pushed the performance even further, reaching **96.99%**. This improvement suggests that the attention mechanism is crucial for improving the model's ability to focus on critical regions in the images, rather than treating all pixel regions equally.

7.4.1 Key Findings:

1. **Attention Mechanism:** The Multi-Head Attention block allowed the model to dynamically weigh different parts of the input data based on their relevance. By learning contextual relationships, the model was able to make better-informed decisions, leading to a substantial improvement in classification performance. This is particularly valuable in medical imaging tasks, where subtle variations in image features can be crucial for accurate diagnosis.
2. **Robustness and Generalization:** The DenseNet121 + Multi-Head Attention model demonstrated enhanced generalization and robustness. This was evident from its ability to achieve higher validation accuracy compared to other models, which did not benefit from attention mechanisms. The ability to focus on the most relevant features allowed the model to adapt better to the complexities of the dataset, ensuring more reliable performance on unseen data.
3. **CNNs as Feature Extractors:** While convolutional neural networks

(CNNs) such as VGG16 and DenseNet121 are strong feature extractors, our results suggest that integrating attention mechanisms can significantly improve their ability to process complex, subtle data. In the case of Parkinson’s disease, where early symptoms are subtle and often difficult to distinguish, the attention mechanism’s ability to highlight important image regions is crucial for improving classification accuracy.

4. **Over-regularization in EfficientNetB0:** EfficientNetB0, despite being a highly optimized model, performed poorly due to over-regularization when applied to the small dataset. This highlights the importance of balancing regularization techniques, especially in medical datasets with limited data. Over-regularization can prevent the model from capturing the finer details necessary for accurate classification.

In conclusion, the integration of DenseNet121 with a Multi-Head Attention mechanism significantly boosted the performance of the Parkinson’s disease classification task. The results affirm that while CNNs are excellent feature extractors, the addition of attention mechanisms can enhance their capabilities, especially for complex medical image analysis tasks, improving both the robustness and generalization of the model.

Chapter 8

CONCLUSION

This study has successfully demonstrated the efficacy of deep learning models in detecting Parkinson's disease using medical imaging data. The comparison of various models revealed that DenseNet121, with its efficient use of deep feature reuse, achieved a high validation accuracy of 96.39%. Furthermore, the integration of a Multi-Head Attention mechanism into DenseNet121 significantly boosted the model's performance, achieving an impressive 96.99% accuracy. This highlights the potential of combining convolutional neural networks (CNNs) with attention mechanisms to improve the classification of subtle and complex medical conditions like Parkinson's disease.

The key findings from the study underscore the importance of model selection in medical image classification tasks. While traditional CNNs such as VGG16 and EfficientNetB0 performed adequately, the DenseNet121 model, with its deeper architecture and advanced feature reuse, demonstrated superior performance. Additionally, the incorporation of a transformer-based attention mechanism allowed the model to focus on important features, making it more robust and capable of generalizing better to unseen data.

The successful application of these deep learning techniques to Parkinson's disease classification suggests that similar approaches can be extended to other medical conditions, potentially aiding in early detection and diagnosis. This work lays the foundation for further exploration into the use of

advanced neural network architectures and attention mechanisms in medical imaging tasks.

8.1 SCOPE OF FURTHER WORK

While the current study has yielded promising results, there are several areas where future research can build upon this work. The following outlines potential directions for further exploration:

8.1.1 Future Direction in the Project

The integration of attention mechanisms into deep learning models has shown significant promise in improving classification accuracy, especially for complex medical image analysis tasks. However, there are several avenues where future research could focus to improve upon this work:

1. **Expansion to Other Medical Datasets:** This study focused specifically on Parkinson’s disease, but the approach could be expanded to include datasets for other neurodegenerative diseases, such as Alzheimer’s disease or multiple sclerosis. Exploring the generalizability of the DenseNet121 + Multi-Head Attention model on these datasets would provide valuable insights into the robustness and versatility of the architecture.

2. **Optimization of Attention Mechanisms:** While the multi-head attention mechanism was effective in improving model performance, there is potential for further optimization. Exploring variations of attention mechanisms, such as self-attention or dynamic attention, could lead to even greater improvements in performance, especially in tasks with subtle image variations.

3. **Incorporating Multi-modal Data:** Future work could explore incorporating multiple modalities of data, such as clinical data (e.g., patient

history, genetic information) along with medical images. The fusion of image data with other patient-related information could lead to more accurate and comprehensive models for disease detection and prognosis.

4. Model Interpretability: One of the major challenges in deep learning, particularly in medical applications, is the lack of interpretability of the models. Future research could focus on improving the transparency of the DenseNet121 + Multi-Head Attention model by using techniques like explainable AI (XAI). This would enable clinicians to understand the decision-making process of the model, making it a more practical tool for real-world medical applications.

5. Real-time Classification: Although this study achieved high accuracy on the validation set, the performance of the model in real-time classification scenarios (such as in a clinical setting) needs further investigation. Research can explore ways to improve the speed of the model's inference, ensuring that it can be applied in practical, time-sensitive environments.

6. Transfer Learning from Larger Datasets: The current model was trained on a relatively small dataset, which may limit its performance and generalization to real-world data. Future work could investigate the use of transfer learning from larger, more diverse datasets to improve the model's robustness and ability to generalize to unseen data.

In conclusion, while this study represents a significant step forward in Parkinson's disease detection, there is ample opportunity for further research to enhance the model's performance, broaden its applicability, and improve its practical utility in clinical environments. Through these future directions, we can continue to push the boundaries of deep learning in medical applications, ultimately improving early diagnosis and treatment of complex diseases.

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